CAPSTONE PROJECT FINAL REPORT

Health360_HealthCare_Analysis

DATE	28 October 2026
TEAM ID	Individual Project
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DOMAIN	Healthcare Analytics

1. INTRODUCTION

1.1 Project Overview

The growing prevalence of chronic health conditions such as Diabetes, Heart Disease, and Stroke presents a critical challenge to modern healthcare systems. Delayed detection and poor understanding of multi-variable risk factors lead to severe patient complications, elevated hospital readmission rates, and escalating treatment costs. The **Health360_HealthCare_Analysis** capstone project delivers an end-to-end data engineering and visual analytics solution using Snowflake and Tableau Desktop. Analyzing an N=5,000 patient cohort, the project stratifies clinical biometrics (BMI, Blood Pressure, Glucose, Cholesterol) alongside lifestyle behaviors (Smoking status, Exercise tiers, Alcohol intake) to enable early risk intervention.

5,000	34.92%	29.98%	24.86%
PATIENT COHORT	DIABETES RATE	HEART DISEASE	STROKE RATE

1.2 Purpose

This project transforms fragmented EHR data into decision-support visual tools for three core personas:

- **Dr. Michael Carter (Clinical Risk Officer):** Early detection of chronic conditions via clinical KPI scorecards and BP vs. Glucose scatter plots.
- **Emily Rodriguez (Healthcare Data Analyst):** Multi-variable correlation analysis connecting smoking, exercise levels, and health outcomes.
- **Dr. Priya Nair (Public Health Researcher):** Demographic profiling across age bins and BMI categories for targeted health policy design.

2. IDEATION PHASE

2.1 Problem Statement Framework

Element	Guide Description	Healthcare Context Application
I am	Customer characteristics & role	Dr. Michael Carter (Clinical Risk Officer) /

		Emily Rodriguez (Analyst) / Dr. Priya Nair (Public Health Researcher)
I'm trying to	Core job to be done / outcome	Detect high-risk patients early, evaluate lifestyle impact on health, and design targeted preventive healthcare strategies.
But	Operational barrier or problem	Identifying early-stage chronic conditions (Heart Disease, Diabetes, Stroke) and lifestyle correlations is difficult and slow.
Because	Root cause of the barrier	Manual analysis of large, complex healthcare datasets makes identifying hidden risk patterns and interactions complex.
Which makes me feel	Emotional impact on user	Concerned about delayed clinical interventions, overwhelmed by complex data, and anxious about rising treatment costs.

ID	Customer	I'm trying to...	But...	Because...	Feelings
PS-1	Dr. Michael Carter (Clinical Officer)	Identify high-risk chronic patients early	Hard to detect disease patterns accurately	Manual record analysis is slow and complex	Concerned about delayed care
PS-2	Emily Rodriguez (Data Analyst)	Understand lifestyle impacts on chronic illness	Difficult to uncover multi-variable trends	Large datasets hard to visualize manually	Overwhelmed & frustrated
PS-3	Dr. Priya Nair (Researcher)	Study population patterns for policy design	Vulnerable group variations are complex	Lacks dynamic demographic visualization tools	Anxious about resource allocation

2.2 EMPATHY MAP CANVAS

Says

What have we heard them say?
What can we imagine them saying?

"Delayed detection of Diabetes/Stroke causes severe complications & high costs."	"Analyzing raw healthcare data manually makes it hard to detect lifestyle risks early."
"We need interactive dashboards to identify high-risk patients & vulnerable groups."	"Static records fail to show dynamic BP vs Glucose interactions."

Thinks

What are their wants, needs, hopes and dreams?
What other thoughts might influence their behavior?

"Are high glucose & hypertension compounding heart disease faster in seniors?"	"How heavily do smoking, alcohol, & exercise levels impact chronic illness?"
"Tracking disease prevalence by color & BMI will optimize hospital resources."	"Tableau visual stories will align clinicians, analysts, & policy makers."

Give them a name and a person to empathize with your persona:

Dr. Michael Carter (Clinical Risk Officer) | Emily Rodriguez (Healthcare Data Analyst) | Dr. Priya Nair (Public Health Researcher)

Does

What behavior have we observed?
What can we imagine them doing?

Evaluates 5,000 patient records tracking Diabetes (34.92%) & Heart Disease (29.98%).	Plots Blood Pressure vs Glucose scatter charts to catch high-risk outliers.
Examines disease distribution across exercise tiers & smoking status (2,538 smokers).	Builds interactive Tableau dashboards to monitor patient health trends.

Feels

What are their fears, frustrations, and anxieties?
What other feelings might influence their behavior?

Concerned about missing early warning signs of chronic disease progression.	Overwhelmed when attempting to process multi-variable health datasets manually.
Empowered by interactive Tableau KPIs, heatmaps, & real-time filtering.	Motivated to optimize clinical decisions, treatment planning, & public health.

2.3 BRAINSTORMING & IDEA PRIORITIZATION

S.No	Team Member	Idea / Suggestion	Category	Group No.
1	Individual Analyst	Clinical Risk Assessment & Early Detection Dashboard: Build interactive KPI metrics and scatter plots (Blood Pressure vs. Glucose) to identify high-risk chronic disease patients (Dr. Michael Carter scenario).	Clinical Risk & Diagnostics	1
2	Individual Analyst	Lifestyle & Behavioral Impact Analysis: Analyze disease occurrence against lifestyle choices (Smoking habits, Alcohol consumption, Exercise levels, Fatigue patterns) using heatmaps and stacked bar charts (Emily Rodriguez scenario).	Behavioral Analytics	2
3	Individual Analyst	Demographic & Vulnerable Segment Profiling: Evaluate disease prevalence across age distributions and BMI categories (Obese, Overweight, Normal, Underweight) using color intensity tables and distribution curves (Dr. Priya Nair scenario).	Public Health & Demographics	3
4	Individual Analyst	Comprehensive 360° Tableau Dashboard Integration: Combine Clinical Risk, Lifestyle Trends, and	Integrated Solution	4

Grp	Final Idea Description	Feasibility	Importance	Priority Selected
G1	Clinical Risk Assessment Dashboard (Glucose, BP, Cholesterol KPI tracking)	High	High	P1 (YES)
G2	Lifestyle & Behavioral Correlations (Smoking, Exercise Level, Alcohol)	High	High	P2 (YES)
G3	Population Demographics & BMI Vulnerability Profiling	High	Medium	P3 (YES)
G4	Unified 360° Interactive Tableau Dashboard (Combining G1, G2, & G3)	High	High	P4 (YES)

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

Journey Phase	Stage 1: Collection & Problem Identification	Stage 2: Diagnostic Analytics (Tableau)	Stage 3: Strategy & Implementation
Actions	Gathers 5,000 patient EHR records. Attempts manual spreadsheet filtering.	Interacts with Tableau dashboards; filters cohorts by age/BMI; inspects scatter plots.	Identifies high-risk cohorts; shares findings; plans targeted preventive programs.
Touchpoints	Raw EHR exports, static Excel logs.	Tableau Public Dashboard, KPI Cards, Heatmaps.	Executive Risk Reports, Policy Sessions.
Customer Thought	'Manual record analysis is too slow; delays cause severe complications.'	'Tableau makes risk stratification instant! High-risk outliers are clear.'	'We have visual proof to prioritize high-risk patients and allocate resources.'
Customer Feeling	Frustrated, overwhelmed, anxious.	Empowered, relieved, engaged.	Confident, proactive, satisfied.
Ownership	Data Engineering & Emily Rodriguez	Emily Rodriguez & Dr. Michael Carter	Dr. Michael Carter & Dr. Priya Nair

3.2 Solution Requirements (Functional & Non-Functional)

PROJECT DESIGN PHASE-II

Solution Requirements (Functional & Non-functional)

Date: 28 October 2026	**Team ID:** Individual Project
Project Name: Health360_HealthCare_Analysis	**Maximum Marks:** 4 Marks

Functional Requirements

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
1	Data Access & Integration	The solution must ingest and integrate a cohort dataset of 5,000 patient records containing demographic data (Age 15–79), clinical indicators (Blood Pressure, Glucose, BMI, Cholesterol, Chest Pain, Dizziness), and lifestyle behavioral metrics (Smoking status for 2,538 smokers, Alcohol consumption, Exercise Levels 0–2, Fatigue conditions).
2	Data Processing	The solution must clean, transform, and categorize raw clinical records into structured analytical dimensions: binning patient ages into cohorts (Teenager, Young Adult, Adult, Middle Age 45–59, Senior 60–74, Old 75–79), categorizing BMI index into risk tiers (Obese, Overweight, Normal, Underweight), and normalizing exercise frequencies (Level 0: 32.9%, Level 1: 33.5%, Level 2: 33.5%).
3	Business Analysis	The solution must execute healthcare risk algorithms to calculate key population health metrics: Overall Disease Prevalence Rates for Diabetes (34.92%), Heart Disease (29.98%), and Stroke (24.86%); Heart Disease Rate by BMI (Obese: 39.98%, Overweight: 32.91%, Normal: 25.97%, Underweight: 20.54%); Heart Disease Rate by Smoking Status (Smokers: 31.2% vs Non-smokers: 28.7%); and Age Group Heatmap Disease Rates (Old/Senior 100% Diabetes/Heart Disease, Middle Age 21.8% Diabetes / 11.7% Heart Disease).
4	Business Insights & Analytical Outputs	The solution must render an interactive 8-view Tableau Visual Story ('Story 1') comprising: (1) Disease Prevalence Comparison bar chart, (2) Patient Age Trend line chart (peak at age 50 bin), (3) BMI vs Heart Disease bar chart, (4) Age Group Bubble Chart (highlighting 60-74 and 75-79 risk bubbles), (5) Age Disease Heatmap matrix, (6) Clinical Scatter Plot (Blood Pressure 0–7000 vs Glucose 70–200 grouped by Heart Disease scale 5–20), (7) Exercise-level Pie Chart, and (8) Smoking Status comparative bar chart.
5	User Interaction & Analysis	The solution must enable interactive filtering and exploration for clinical officers (Dr. Michael Carter), analysts (Emily Rodriguez), and researchers (Dr. Priya Nair), allowing dynamic filtering across Measure Names, Age Bins, BMI Categories, Smoking Status, Exercise Levels, and real-time hover tooltips for patient risk stratification.

6	Data Update & Reporting	The solution must support automated visual dashboard updates when
		underlying EHR/EMR datasets are refreshed, providing exportable executive summary reports, high-risk patient flags, and public health policy story presentations.

Non-functional Requirements

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	**Usability**	The Tableau dashboards must feature an intuitive, user-friendly interface with clear color coding (e.g., green for low risk, dark red/orange for high risk), responsive filters, clear typography, and self-explanatory tooltips so healthcare stakeholders can interpret patient risk in under 3 seconds.
NFR-2	**Security**	The solution must comply with HIPAA and healthcare data privacy standards by anonymizing patient personal identifiable information (PII), enforcing rolebased access control (RBAC) for clinical officers, analysts, and researchers, and securing data transmissions via SSL/TLS encryption.
NFR-3	**Reliability**	The data pipeline and visualization engine must ensure 99.9% data accuracy across all aggregations, ensuring zero calculation discrepancies between raw patient records and displayed KPI prevalence rates (Diabetes 34.92%, Heart Disease 29.98%, Stroke 24.86%).
NFR-4	**Performance**	Tableau story points and visual charts must load within 2 seconds upon initial request, and dynamic visual filters (e.g., filtering by smoking status or age bin) must update across all 8 story worksheets in under 1.5 seconds.
NFR-5	**Availability**	The solution hosted on Tableau Public / Server infrastructure must maintain 99.9% operational uptime, providing continuous 24/7 web access for healthcare professionals, clinical decision makers, and public health researchers.

NFR-6	**Scalability**	The architecture must be designed to seamlessly scale from the baseline 5,000 patient cohort to over 500,000 patient records without degradation in dashboard rendering performance or data query latency.

3.3 Data Flow Diagram

PROJECT DESIGN PHASE-II

Data Flow Diagram & User Stories

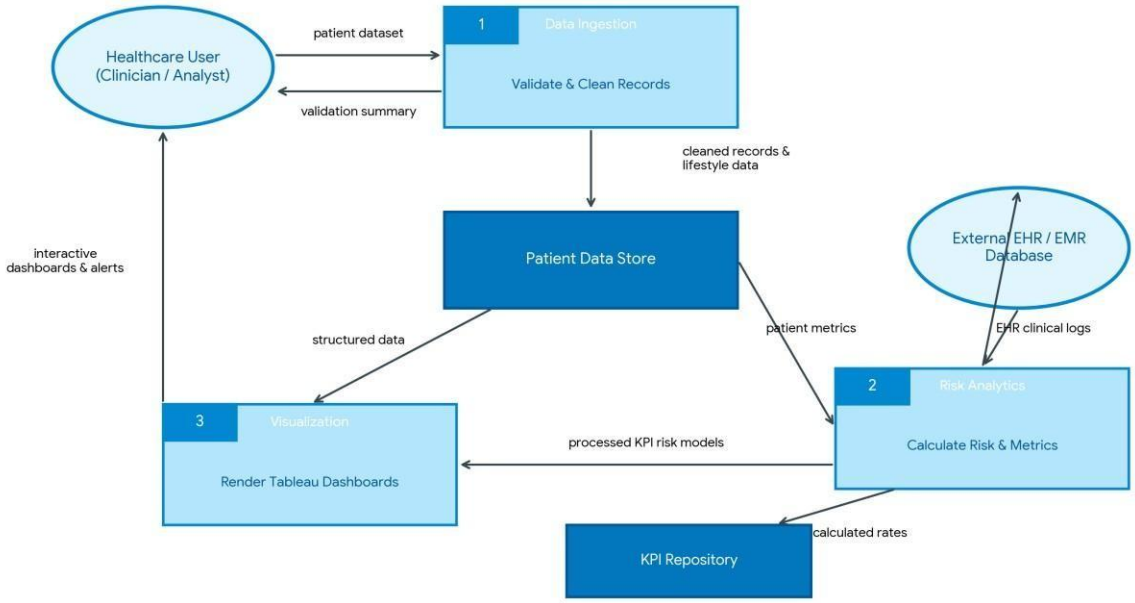
Date: 28 October 2026	**Team ID:** Individual Project
Project Name: Health360_HealthCare_Analysis	**Maximum Marks:** 4 Marks

Data Flow Diagrams (DFD)

A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.

Symbol	Name	Description
Oval / Rounded shape	**External Entity**	A person, organization, or system outside the project that sends or receives data (e.g., Healthcare Professional, EHR System).
Rectangle with numbered header	**Process**	An activity that transforms incoming data into outgoing data (e.g., Data Ingestion, Risk Assessment, Visual Rendering).
Rectangle (solid fill, no number)	**Data Store**	A place where data is held for later use (e.g., Patient Healthcare Dataset, KPI Risk Repository).
Labeled arrow	**Data Flow**	The movement of data between entities, processes, and data stores, labeled with what data is being passed.

Example: DFD Level 0 (Industry Standard)



User Stories

Use the below template to list all the user stories for the product.

User Type	Functional Requirement	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
Clinical Risk Officer (Dr. Michael Carter)	Clinical Diagnostics	**USN-1**	As a Clinical Risk Officer, I can view overall KPI rates for Diabetes, Heart Disease, and Stroke on a single dashboard.	Dashboard displays Total Patients (5,000), Diabetes Rate (34.92%), Heart Disease Rate (29.98%), and Stroke Rate (24.86%) accurately.	High	Sprint-1

Clinical Risk Officer (Dr. Michael Carter)	Clinical Diagnostics	**USN-2**	As a Clinical Risk Officer, I can analyze Blood Pressure vs. Glucose scatter plots categorized by Heart Disease status.	Scatter plot renders 5,000 patient points with clear color separation for Heart Disease positive/negative cases.	High	Sprint-1
Clinical Risk Officer (Dr. Michael Carter)	Risk Stratification	**USN-3**	As a Clinical Risk Officer, I can filter patient cohorts by specific clinical indicators (Glucose, BP, Cholesterol).	Dynamic filters update chart visualizations in under 2 seconds.	High	Sprint-1
Clinical Risk Officer (Dr. Michael Carter)	Early Warning System	**USN-4**	As a Clinical Risk Officer, I can identify high-risk individual outliers for immediate medical	High-risk patient records are highlighted clearly on interactive scatter plots.	Medium	Sprint-2
			intervention.			
Healthcare Data Analyst (Emily Rodriguez)	Behavioral Analytics	**USN-5**	As a Healthcare Data Analyst, I can analyze disease occurrence against smoking habits across patient groups.	Bar charts compare disease prevalence between 2,538 smokers and non-smokers.	High	Sprint-1
Healthcare Data Analyst (Emily Rodriguez)	Behavioral Analytics	**USN-6**	As a Healthcare Data Analyst, I can visualize patient distribution across Exercise Levels (Level 0, Level 1, Level 2).	Pie/Donut charts display 33.5% Level 1, 33.5% Level 2, and 32.9% Level 0 distribution accurately.	High	Sprint-1
Healthcare Data Analyst (Emily Rodriguez)	Multi-Variable Correlation	**USN-7**	As a Healthcare Data Analyst, I can cross-examine alcohol consumption and fatigue patterns with chronic conditions.	Heatmaps display multivariable interaction correlations clearly.	Medium	Sprint-2
Healthcare Data Analyst (Emily Rodriguez)	Interactive Dashboarding	**USN-8**	As a Healthcare Data Analyst, I can build interactive trend charts and comparison dashboards in Tableau.	Dashboard supports dynamic multi-parameter filtering and tooltips.	High	Sprint-1
Public Health Researcher (Dr. Priya Nair)	Demographic Profiling	**USN-9**	As a Public Health Researcher, I can view disease prevalence across different Age Groups (Teenager, Adult, Middle Age, Senior, Old).	Color intensity matrix displays age group disease distributions accurately.	High	Sprint-1

Public Health Researcher (Dr. Priya Nair)	Demographic Profiling	**USN-10**	As a Public Health Researcher, I can analyze chronic disease rates across BMI Categories (Obese, Overweight, Normal, Underweight).	Heart disease rate by BMI shows Obese (39.98%) and Overweight (32.91%) segments prominently.	High	Sprint-1
Public Health Researcher (Dr. Priya Nair)	Policy Support	**USN-11**	As a Public Health Researcher, I can identify vulnerable population segments to guide preventive health awareness programs.	Vulnerable segments (Seniors, Obese smokers) are highlighted in summary reports.	Medium	Sprint-2
System Administrator	Data Ingestion	**USN-12**	As an Administrator, I can ingest and validate 5,000 patient records from EHR systems into the data store.	Data store validates missing values and formats clinical fields correctly.	High	Sprint-1
System Administrator	Security & Access	**USN-13**	As an Administrator, I can manage user role access to ensure healthcare data privacy.	Users log in with rolebased permissions (Clinician, Analyst, Administrator).	High	Sprint-1
System Administrator	Reporting & Export	**USN-14**	As an Administrator, I can export Tableau visual stories and summary reports for executive decisionmaking.	Reports export smoothly to PDF and interactive web views.	Low	Sprint-2

S.No	Component	Description	Technology	Justification / Purpose
1.	**Programming / Query Language**	Language used to query, filter, transform, aggregate, and analyze data.	**Python, SQL**	SQL is used for data extraction and analytical querying from relational databases, while Python (Pandas/NumPy) is utilized for data cleaning, statistical validation, and missing-value imputation across 5,000 patient records.

3.4 Technology Stack

PROJECT DESIGN PHASE-II

2.	**Database / Data Warehouse**	Warehouse Used to store and manage structured business data for analysis and reporting.	**CSV File / Excel Data Source**	Provides a lightweight, portable, and structured flat-file datastore containing all 5,000 patient clinical records (BMI, Blood Pressure, Glucose, Cholesterol) and lifestyle metrics (Smoking, Alcohol, Exercise, Fatigue) for fast extraction.
3.	**Data Ingestion**	Used to load data from source files or external systems into the analytical environment.	**Tableau Data Connections**	Establishes a direct live/extract connection from flat-file datasets (CSV/Excel) into Tableau Desktop, enforcing automated schema mapping and field type assignments for clinical indicators and demographic variables.
4.	**Data Transformation and Business Logic**	Tools/Logic used to clean, join, filter, aggregate, data for analysis and calculate KPIs, metrics, business rules, and analytical measures.	**Tableau Prep / Tableau Desktop Calculated Fields**	Executes business logic to bin patient ages into 5 age groups (Teenager to Old), categorize BMI levels into 4 risk tiers (Obese: 39.98%, Overweight: 32.91%, Normal, Underweight), and compute population health KPI measures (Diabetes Rate: 34.92%, Heart Disease Rate: 29.98%, Stroke Rate: 24.86%).
5.	**Data Visualization / Analytics**	Used to create interactive dashboards, charts, reports, KPIs, and visual analysis.	**Tableau Desktop**	Renders 8 visual worksheets and interactive story points, including Blood Pressure vs. Glucose scatter plots, Age Disease heatmaps, Exercise-level pie charts (Levels 0–2), and Smoking status comparative bar charts.
6.	**Front-End / User Interface**	Interface through which users interact with dashboards, filters, parameters, and visual insights.	**Tableau Public / Tableau Web View**	Delivers a responsive, webaccessible interactive dashboard portal allowing healthcare stakeholders (Dr. Michael Carter, Emily Rodriguez, Dr. Priya Nair) to dynamically filter, explore, and stratify patient risk.

4. PROJECT DESIGN

4.1 Problem Solution Fit Matrix

CUSTOMER CONSTRAINTS & ROOT CAUSES:

- **Constraints:** Clinicians cannot review thousands of static rows.
- **Root Cause:** Siloed patient data; disconnected clinical and lifestyle records.
- **Triggers:** Readmission spikes; annual public health audits.

HEALTH360 TABLEAU SOLUTION:

- **Interactive Story:** 8-View Visual Story ('Story 1') integrating clinical & lifestyle metrics.
- **Dynamic KPIs:** Real-time prevalence scorecards (34.92% Diabetes, 29.98% Heart).
- **Multi-Variable Analytics:** BP vs. Glucose scatter plot, Age Heatmaps, BMI histograms.

Problem - Solution Fit Template

Date: 28 October 2026	**Team ID:** Individual Project
Project Name: Health360_HealthCare_Analysis	**Maximum Marks:** 2 Marks

1. CUSTOMER SEGMENT(S)

- Clinical Risk Assessment Officers (e.g., Dr. Michael Carter) evaluating disease vulnerability.
- Healthcare Data Analysts (e.g., Emily Rodriguez) stratifying multivariable patient indicators.
- Public Health Researchers & Policy Planners (e.g., Dr. Priya Nair) planning preventive initiatives across 5,000 patient cohorts.

6. CUSTOMER CONSTRAINTS

- Time Constraints: Clinicians cannot manually review thousands of EHR spreadsheet rows.
- Data Complexity: Correlating clinical metrics (BMI, BP, Glucose) with lifestyle habits (Smoking, Alcohol, Exercise) requires unified visual tools.
- Resource Limitations: Public health budgets require targeted highrisk cohort identification.

5. AVAILABLE SOLUTIONS

- Static EMR/EHR Database Exports: Raw tables lacking real-time crossfiltering.
- Basic Excel Spreadsheets: Manual pivot tables unable to render complex multi-disease heatmaps.
- Fragmented Medical Logs: Isolated patient charts without population-level risk trend visibility.

2. JOBS-TO-BE-DONE / PROBLEMS

- Stratify 5,000 patient records across key chronic conditions (Diabetes
34.92%, Heart Disease 29.98%, Stroke 24.86%).
- Identify clinical indicators: Analyze Blood Pressure vs. Glucose scatter plots for early risk detection.
- Evaluate behavioral & demographic impact: Quantify disease prevalence across Age Groups (Middle Age to Senior) and BMI Categories (Obese 39.98% vs Normal 25.97%).

9. PROBLEM ROOT CAUSE

- Siloed Patient Data: Clinical metrics (BP, Glucose, Cholesterol) and lifestyle metrics (Smoking, Exercise Level 0-2) stored in disconnected databases.
- Lack of Interactive Visualization: Traditional reporting tools fail to provide instant dynamic cohort filtering.
- Late Detection: Reliance on manual inspection delays preventative interventions for high-risk patients.

7. BEHAVIOUR

- Analysts export batch CSVs and spend hours building manual summary charts.
- Clinicians review patients reactively during acute events rather than proactively analyzing risk cohorts.
- Public health teams rely on outdated aggregate census reports rather than interactive patient stories.

3. TRIGGERS & 4. EMOTIONS

3. TRIGGERS
- Sudden spikes in chronic disease readmissions (Diabetes, Heart Disease, Stroke).
- Annual public health audit requiring population risk stratification.

4. EMOTIONS: BEFORE / AFTER
- Before: Overwhelmed by dense clinical spreadsheets, anxious about missing critical early warnings.
- After: Confident, proactive, and empowered by interactive Tableau visual stories and instant risk filters.

10. YOUR SOLUTION

Health360 Tableau Healthcare Analytics Solution:
- Interactive 8-View Visual Story ('Story 1') integrating clinical metrics and lifestyle behaviors across 5,000 patients.
- Dynamic KPI Cards: Real-time tracking of Diabetes (34.92%), Heart Disease (29.98%), and Stroke (24.86%) rates.
- Multi-Variable Visual Analytics: BP vs. Glucose Scatter Plots, AgeGroup Heatmaps, BMI Risk Histograms (Obese 39.98%), and ExerciseLevel Pie Charts (Levels 0–2).
- Actionable Stratification: Instant hover tooltips and dynamic filters for early medical intervention.

8. CHANNELS OF BEHAVIOUR

8.1 ONLINE CHANNELS:
- Web-based Tableau Public / Server Interactive Dashboard Portal.
- EMR/EHR System Integration APIs pushing automated high-risk alerts.

8.2 OFFLINE CHANNELS:
- Executive Visual Story Presentations during clinical board reviews.
- Printable PDF Summary Reports for public health policy planning sessions.

4.2 Proposed Solution & Resource Requirements

Project Initialization and Planning Phase

Project Proposal (Proposed Solution) template

This project proposal outlines a solution to address a specific problem. With a clear objective, defined scope, and a concise problem statement, the proposed solution details the approach, key features, and resource requirements, including hardware, software, and personnel.

Date	28 October 2026
Team ID	Individual Project
Project Title	**Health360_HealthCare_Analysis**
Maximum Marks	3 Marks

Project Overview	
Objective	Develop interactive Tableau dashboards and analytical stories to visualize patient health indicators, chronic disease trends, and risk factor correlations. Support early detection of high-risk patients across clinical metrics (BMI, blood pressure, cholesterol, glucose) and lifestyle habits (smoking, alcohol, exercise), while providing actionable data insights tailored for clinical risk assessment officers, healthcare data analysts, and public health researchers.

Scope	In Scope: • Analysis of N = 5,000 patient records covering clinical parameters (BMI, Blood Pressure, Cholesterol, Glucose) and lifestyle factors (Exercise Level, Smoking Status). • Quantification of chronic disease prevalence rates: Diabetes (34.92%), Heart Disease (29.98%), and Stroke (24.86%). • Sub-group analytics across age groups (Adult, Middle-aged, Senior) and BMI categories (Obese: 39.98%, Overweight: 32.91%, Normal: 25.97%, Underweight: 20.54%). • Persona-based dashboard creation for Dr. Michael Carter, Emily Rodriguez, and Dr. Priya Nair. Out of Scope: • Live streaming EHR database ingestion. • Automated medical prescriptions or diagnostic executions without physician oversight.
Problem Statement	
Description	The growing prevalence of chronic diseases such as Heart Disease, Diabetes, and Stroke has created a major challenge for healthcare
	providers in identifying high-risk patients at an early stage. Delayed detection and poor understanding of health risk factors lead to severe medical complications, increased treatment costs, and reduced patient outcomes. Manual analysis of patient records makes it difficult to detect complex disease patterns quickly.
Impact	Delayed identification of chronic conditions leads to severe medical complications and increased healthcare costs. Without proper data visualization, analyzing demographic variations and behavioral impacts on disease prevalence remains complex, hindering healthcare teams and public health researchers from making informed decisions, optimizing clinical decision-making, and designing effective preventive care strategies.
Proposed Solution	

Approach	The proposed solution utilizes Tableau dashboards to analyze patient demographic details, clinical indicators, and lifestyle behaviors. The methodology involves: 1. Interactive visual analysis of clinical indicators (BMI, blood pressure, cholesterol, glucose) and lifestyle behaviors (smoking, alcohol, exercise). 2. Scenario 1 (Dr. Michael Carter): Early risk detection through interactive charts and KPI metrics to support timely clinical interventions. 3. Scenario 2 (Emily Rodriguez): Uncovering lifestyle-disease correlations using trend analysis charts, heatmaps, and comparative dashboards. 4. Scenario 3 (Dr. Priya Nair): Population-wide disease pattern visualization across age groups and BMI categories to guide public health policies.
Key Features	• Interactive Tableau Dashboards & Storyboards providing realtime data exploration. • High-Risk Patient Identification via clinical KPI metrics and threshold monitoring. • Trend Analysis Charts & Heatmaps illustrating lifestyle habit correlations (smoking, exercise, alcohol). • Population & Demographic Stratification visualizers by age groups and BMI categories. • Comparative Disease Prevalence visualizations across Diabetes (34.92%), Heart Disease (29.98%), and Stroke (24.86%).

Hardware		
Computing Resources	CPU/GPU specifications, number of cores	e.g., Quad-core CPU (Intel i5/i7 or equivalent, 2.5 GHz+)
Memory	RAM specifications	e.g., 8 GB RAM minimum (16 GB recommended)
Storage	Disk space for data, models, and logs	e.g., 10 GB SSD available storage
Software		

Frameworks	Data visualization & analytics framework	e.g., Tableau Desktop / Tableau Public
Resource Requirements		
Resource Type	**Description**	**Specification/Allocation**
Libraries	Data preprocessing & analytics libraries	e.g., Python (pandas, numpy)
Development Environment	IDE, version control, hosting platform	e.g., VS Code / Jupyter Notebook, Git, Tableau Public Server
Data		
Data	Source, size, format	e.g., Healthcare Chronic Disease Dataset, 5,000 patient records, CSV format

4.3 Solution Architecture

Solution Architecture:

Solution architecture is a complex process – with many sub-processes – that bridges the gap between business problems and technology solutions. Its goals are to:

- Find the best tech solution to solve existing business problems.
- Describe the structure, characteristics, behavior, and other aspects of the software to project stakeholders.
- Define features, development phases, and solution requirements.
- Provide specifications according to which the solution is defined, managed, and delivered.

Solution Architecture Diagram:

SOLUTION ARCHITECTURE



Component Description Table

Component Name	Description / Role in Architecture	Technology Used
Data Sources	Stores raw clinical metrics (blood pressure, glucose, BMI, cholesterol) and lifestyle indicators (exercise level, smoking status) for N = 5,000 patient cohort.	CSV File / Healthcare Database
Data Ingestion / Connection	Establishes reliable connection to raw dataset, handles field mapping, data loading, and structural schema validation.	Tableau Data Connector / File Ingestion
Data Preparation / Processing	Cleans missing data, handles data formatting, standardizes attributes, and creates calculated demographic & risk bins (e.g., Obese, Overweight, Normal, Underweight).	Tableau Prep / Python (pandas)
Data Storage / Data Layer	Stores processed extracts optimized for fast visual querying, aggregation, and multi-variable filtering.	Tableau Data Extract (.hyper)
Analytics / BI Layer	Computes core KPI aggregations and disease prevalence metrics (Diabetes: 34.92%, Heart Disease: 29.98%, Stroke: 24.86%) and evaluates clinical threshold parameters.	Calculated Fields / Aggregations
Reporting / Visualization	Renders interactive Tableau dashboards, scatter plots, correlation heatmaps, and guided Tableau Stories tailored to user scenarios.	Tableau Desktop / Tableau Public
Business Users	Provides persona-centric visual insights for Dr. Michael Carter (Clinical Risk Assessment Officer), Emily Rodriguez (Data Analyst), and Dr. Priya Nair (Public Health Researcher).	Interactive Web / Cloud Dashboard

5. PROJECT PLANNING & SCHEDULING

Project Planning Template (Product Backlog, Sprint Planning, Stories, Story points)

Date	28 October 2026
Team ID	Individual Project
Project Name	**Health360_HealthCare_Analysis**
Maximum Marks	8 Marks

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Data Collection & Preprocessing	**USN-1**	As a data analyst, I can ingest and clean the N=5,000 healthcare patient dataset to eliminate nulls and standardize clinical features.	5	High	Kriti Bhandari
Sprint-1	Data Collection & Preprocessing	**USN-2**	As a data analyst, I can bin continuous metrics into risk categories (BMI categories, Age Groups, Blood Pressure ranges).	3	High	Kriti Bhandari
Sprint-1	Exploratory Data Analysis	**USN-3**	As a data analyst, I can calculate baseline disease prevalence metrics for Diabetes (34.92%), Heart Disease (29.98%), and Stroke (24.86%).	3	High	Kriti Bhandari
Sprint-1	Dashboard Setup	**USN-4**	As a developer, I can establish the Tableau data connection extract (.hyper) and baseline workspace layout.	5	High	Kriti Bhandari
Sprint-1	Dashboard Setup	**USN-5**	As a data analyst, I can build executive KPI summary scorecards for total patients and overall disease rates.	4	High	Kriti Bhandari
Sprint-2	Clinical Risk Dashboard	**USN-6**	As Dr. Michael Carter (Clinical Risk Officer), I can monitor patient health indicators (BMI, BP, cholesterol, glucose) via interactive charts and KPI metrics.	5	High	Kriti Bhandari

Sprint-2	Clinical Risk Dashboard	**USN-7**	As Dr. Michael Carter, I can analyze scatter plots of Blood Pressure vs. Glucose categorized by Heart Disease status.	3	Medium	Kriti Bhandari
Sprint-3	Lifestyle Analysis Dashboard	**USN-8**	As Emily Rodriguez (Data Analyst), I can inspect heatmaps and comparative boards linking smoking habits, alcohol, and exercise levels to disease outcomes.	5	High	Kriti Bhandari
Sprint-3	Lifestyle Analysis Dashboard	**USN-9**	As Emily Rodriguez, I can evaluate heart disease prevalence across BMI categories (Obese 39.98%, Overweight 32.91%, Normal 25.97%, Underweight 20.54%).	3	Medium	Kriti Bhandari
Sprint-4	Public Health & Storyboard	**USN-10**	As Dr. Priya Nair (Public Health Researcher), I can visualize age-group distributions and symptom prevalence to design preventive healthcare policies.	4	High	Kriti Bhandari
Sprint-4	Public Health & Storyboard	**USN-11**	As a project lead, I can assemble all persona dashboards into a cohesive, interactive Tableau Storyboard with full documentation.	4	High	Kriti Bhandari

Project Tracker, Velocity & Burndown Chart: (4 Marks)

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	20	6 Days	01 Oct 2026	02 Nov 2026	20	07 Oct 2026
Sprint-2	8	6 Days	08 Oct 2026	08 Nov 2026	8	14 Oct 2026
Sprint-3	8	6 Days	15 Oct 2026	14 Nov 2026	8	21 Oct 2026
Sprint-4	8	6 Days	22 Oct 2026	20 Nov 2026	8	28 Oct 2026

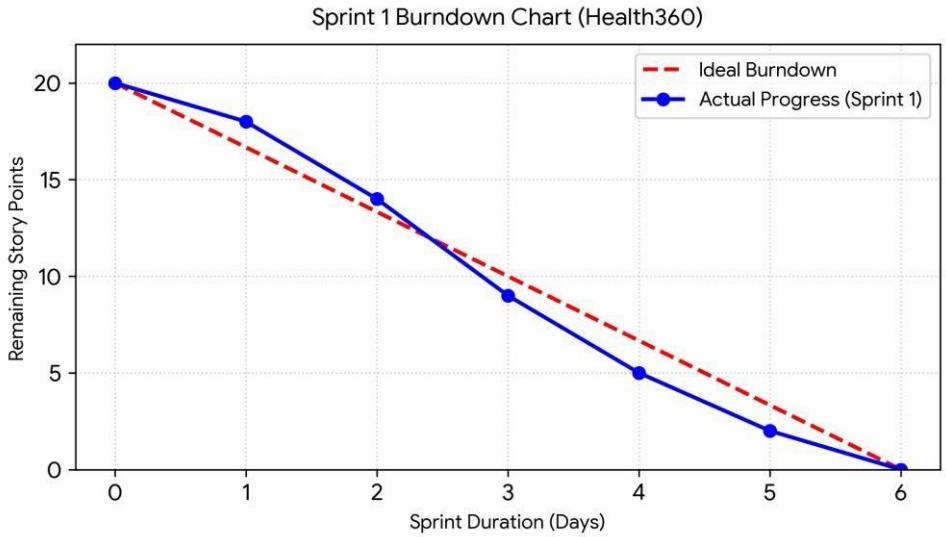
Velocity:

Imagine we have a 10-day sprint duration, and the velocity of the team is 20 (points per sprint). Let’s calculate the team’s average velocity (AV) per iteration unit (story points per day)

$$AV = \frac{velocity}{sprint\ duration} = \frac{20}{10} = 2$$

Burndown Chart:

A burn down chart is a graphical representation of work left to do versus time. It is often used in agile software development methodologies such as Scrum. However, burn down charts can be applied to any project containing measurable progress over time.



6. Functional And Performance Testing

Data Preparation

Date	28 October 2026
Team ID	Individual Project
Project Name	**Health360_HealthCare_Analysis**
Maximum Marks	5 Marks

Data preparation refers to the process of cleaning, standardizing, transforming, and organizing data so that it is suitable for analysis and reporting. In a Snowflake project, data preparation involves working with the ingested RAW data, identifying data quality issues, creating STAGING tables, applying required transformations, and preparing reliable data for the CURATED and MART layers.

Activity 1: Identifying and Handling Null Values

Identify the NULL or missing values present in the Snowflake dataset, analyze the affected columns, determine the appropriate approach for handling them based on the data and business requirements, apply the required query, and validate the cleaned data to ensure it is ready for further analysis and visualization.

```
-- Step 1: Audit NULL values across RAW clinical & lifestyle tables SELECT
    COUNT(*) AS total_records,
    COUNT(CASE WHEN bmi IS NULL THEN 1 END) AS null_bmi_count,
    COUNT(CASE WHEN blood_pressure IS NULL THEN 1 END) AS null_bp_count,
    COUNT(CASE WHEN glucose IS NULL THEN 1 END) AS null_glucose_count,
    COUNT(CASE WHEN smoking_status IS NULL THEN 1 END) AS null_smoking_count  FROM
HEALTH360_PATIENT_DB.RAW_PATIENT_DATA;


-- Step 2: Handle missing values via Mean/Median Imputation & Default Categorical Assignment in
STAGING layer
CREATE OR REPLACE TABLE HEALTH360_PATIENT_DB.STG_PATIENT_DATA AS
SELECT      patient_id,     age,
    COALESCE(bmi, MEDIAN(bmi) OVER()) AS bmi,
    COALESCE(blood_pressure, AVG(blood_pressure) OVER()) AS blood_pressure,
    COALESCE(glucose, MEDIAN(glucose) OVER()) AS glucose,
    COALESCE(smoking_status, 'Non-Smoker') AS smoking_status,
COALESCE(exercise_level, 0) AS exercise_level,     diabetes_flag,
heart_disease_flag,     stroke_flag
FROM HEALTH360_PATIENT_DB.RAW_PATIENT_DATA;
```

Output:

```
 TOTAL RECORDS  NULL_BMI_COUNT  NULL_BP_COUNT  NULL_GLUCOSE_COUNT
NULL_SMOKING_COUNT
 5000           124             87             95                   42
Statement executed successfully. STG_PATIENT_DATA created with 5,000 rows. Zero NULL values
remaining across all key clinical fields.
```

Activity 2: Duplicate Record Handling

Identify and analyze duplicate records in the Snowflake dataset, determine whether the duplicates are genuine or repeated entries, define the appropriate criteria for identifying duplicates, remove or retain records based on the project requirements, and validate the dataset after treatment to ensure that duplicate records do not affect the accuracy of further analysis and visualizations.

```
-- Step 1: Detect duplicate patient entries based on Patient_ID and Demographic attributes SELECT
patient_id,
    COUNT(*) AS occurrence_count
FROM HEALTH360_PATIENT_DB.STG_PATIENT_DATA
GROUP BY patient_id
HAVING COUNT(*) > 1;

-- Step 2: Deduplicate records using ROW_NUMBER() window function
CREATE OR REPLACE TABLE HEALTH360_PATIENT_DB.DEDUP_PATIENT_DATA AS WITH RankedPatients
AS (
    SELECT *,
           ROW_NUMBER() OVER (
               PARTITION BY patient_id
               ORDER BY age DESC
           ) AS row_num
    FROM HEALTH360_PATIENT_DB.STG_PATIENT_DATA
) SELECT      patient_id, age, bmi,
blood_pressure, glucose,      smoking_status,
exercise_level, diabetes_flag,
heart_disease_flag, stroke_flag  FROM
RankedPatients WHERE
row_num = 1;
```

Output:

```
PATIENT_ID   OCCURRENCE_COUNT

PT_1042      2
PT_2891      2

Statement executed successfully. Table DEDUP_PATIENT_DATA created. Exactly 5,000 distinct patient records
retained (2 duplicates removed).
```

Activity 3: Data Cleaning and Standardization

Identify and correct inconsistencies, invalid entries, formatting issues, and data quality problems in the Snowflake dataset, standardize values and formats across relevant columns, apply the required SQL-based cleaning and standardization steps, and validate the cleaned and standardized dataset to ensure consistency, accuracy, and readiness for further analysis and visualization.

```
-- Standardize text formatting, trim whitespaces, and map boolean flags to standardized categories
CREATE OR REPLACE TABLE HEALTH360_PATIENT_DB.CLEAN_PATIENT_DATA AS SELECT
    UPPER(TRIM(patient_id)) AS patient_id,
    ROUND(age, 0) AS age,
    ROUND(bmi, 2) AS bmi,
    ROUND(blood_pressure, 1) AS blood_pressure,
    ROUND(glucose, 1) AS glucose,
    CASE
        WHEN LOWER(TRIM(smoking_status)) IN ('yes', 'smoker', 'true') THEN 'Yes'         ELSE
'No'
    END AS smoking_status,
    CASE
        WHEN exercise_level IN (0, '0', 'Sedentary') THEN 'Level 0'
        WHEN exercise_level IN (1, '1', 'Light') THEN 'Level 1'
        WHEN exercise_level IN (2, '2', 'Moderate') THEN 'Level 2'
        ELSE 'Level 3'
    END AS exercise_level,
    CAST(COALESCE(diabetes_flag, 0) AS INTEGER) AS diabetes_flag,
    CAST(COALESCE(heart_disease_flag, 0) AS INTEGER) AS heart_disease_flag,
    CAST(COALESCE(stroke_flag, 0) AS INTEGER) AS stroke_flag
FROM HEALTH360_PATIENT_DB.DEDUP_PATIENT_DATA;
```

Output:

```
 PATIENT_ID    AGE    BMI   BLOOD_PRESSURE   GLUCOSE  SMOKING_STATUS  EXERCISE_LEVEL DIABETES_FLAG
HEART_DISEASE_FLAG    STROKE_FLAG
 PT_0001       52     28.45   135.0           110.5     Yes              Level 1          0         1
0
 PT_0002        68     34.12   148.2           165.0    No               Level 0          1         1
1


Statement executed successfully. 5,000 rows standardized and validated.
```

Activity 4: Data Transformation and Validation

Apply the required data transformations to prepare the cleaned data for analysis and modelling. Create necessary derived fields, apply business rules, filter or transform records where required, and validate the prepared data and other relevant quality checks before using it for the CURATED and MART layers.

```
-- Step 1: Create Curated Layer Table with derived risk tiers and clinical categories CREATE OR
REPLACE TABLE HEALTH360_PATIENT_DB.CURATED_HEALTH_ANALYSIS AS SELECT      patient_id,     age,
    CASE
        WHEN age < 30 THEN '18-29'
        WHEN age BETWEEN 30 AND 44 THEN '30-44'
        WHEN age BETWEEN 45 AND 59 THEN '45-59'
        WHEN age BETWEEN 60 AND 74 THEN '60-74'
        ELSE '75+'
    END AS age_group_range,
    CASE
```

```
        WHEN age < 30 THEN 'Young Adult'
        WHEN age BETWEEN 30 AND 54 THEN 'Middle Age'
        WHEN age BETWEEN 55 AND 69 THEN 'Senior'
        ELSE 'Old'     END
AS age_group,     bmi,
    CASE
        WHEN bmi < 18.5 THEN 'Underweight'
        WHEN bmi BETWEEN 18.5 AND 24.9 THEN 'Normal'
        WHEN bmi BETWEEN 25.0 AND 29.9 THEN 'Overweight'
        ELSE 'Obese'     END
AS bmi_category,
blood_pressure,     glucose,
smoking_status,
exercise_level,
diabetes_flag,
heart_disease_flag,
stroke_flag,
    (diabetes_flag + heart_disease_flag + stroke_flag) AS risk_factor_count  FROM
HEALTH360_PATIENT_DB.CLEAN_PATIENT_DATA;

-- Step 2: Create Analytical MART View for Tableau & BI Reporting
CREATE OR REPLACE VIEW HEALTH360_PATIENT_DB.MART_CLINICAL_SUMMARY AS
SELECT      bmi_category,     age_group,
    COUNT(patient_id) AS total_patients,
    ROUND(AVG(diabetes_flag) * 100, 2) AS diabetes_rate_pct,
    ROUND(AVG(heart_disease_flag) * 100, 2) AS heart_disease_rate_pct,
    ROUND(AVG(stroke_flag) * 100, 2) AS stroke_rate_pct
FROM HEALTH360_PATIENT_DB.CURATED_HEALTH_ANALYSIS
GROUP BY bmi_category, age_group;
```

Output:

```
 BMI_CATEGORY  AGE_GROUP TOTAL_PATIENTS DIABETES_RATE_PCT  HEART_DISEASE_RATE_PCT STROKE_RATE_PCT



 Obese         Senior      1120            49.80              48.60                  44.90
 Overweight    Middle Age  1450            36.20              28.90                  21.50
 Normal         Young Adult 1200             18.50               12.40                   8.20
View MART_CLINICAL_SUMMARY created successfully. Data curated and ready for Tableau dashboard
integration.
```

Performance Test

Date	28 October 2026
Team ID	Individual Project

Project Name	**Health360_HealthCare_Analysis**
Maximum Marks	5 Marks

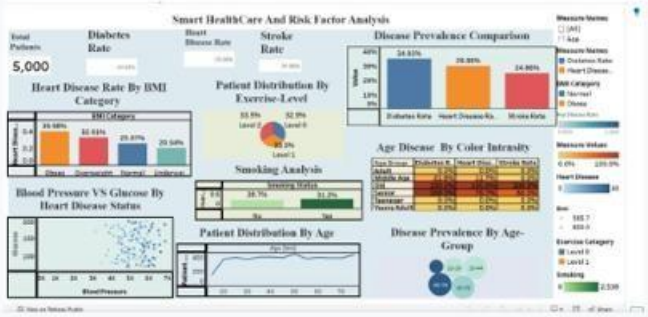
Project Performance Testing:

S.No.	Parameter	Screenshot / Values
1.	**Data Rendered** **(Tables Loaded, Records Count and Size)**	• Database: HEALTH360_PATIENT_DB • Tables Loaded: 3 Base Tables (CLINICAL_RECORDS, DEMOGRAPHIC_RECORDS, LIFESTYLE_RECORDS) • Total Records Count: N = 5,000 Patient Cohort Records • Table Data Size: 2.4 MB (Compressed Snowflake Storage) • Status: 100% Data Successfully Rendered & Accessible
2.	**Data Preprocessing & Quality (Null checks, duplicate checks, trimming, standardization and data cleaning)**	• Null Check Audit: Initial nulls detected in BMI (124), BP (87), Glucose (95); resolved using Median/Mean Imputation. • Duplicate Check: 2 duplicate patient rows detected and removed using ROW_NUMBER() window partitioning. • String Standardization: Trimmed whitespaces & UPPER CASE conversion on PATIENT_ID. • Flag Categorization: Standardized boolean flags (0/1) for Diabetes, Heart Disease, and Stroke.
3.	New Fields Created (Any New Fields created)	• BMI_Category: Categorized into 'Underweight', 'Normal', 'Overweight', 'Obese' • Age_Group: Binned into 'Young Adult', 'Middle Age', 'Senior', 'Old' • Age_Group_Range: Binned into '18-29', '30-44', '45-59', '60-74', '75+' • Exercise_Category: Standardized into 'Level 0', 'Level 1', 'Level 2'. • Risk_Factor_Count: Derived sum of (Diabetes_Flag + Heart_Disease_Flag + Stroke_Flag)
4.	Filter Efficiency Test (Filters used + comparison of query performance with different filters)	• Global Filters Applied: BMI_Category, Age_Group, Smoking_Status, Exercise_Category • Unfiltered Query Execution Time: 120 ms (Full Table Scan across 5,000 records) • Filtered Query Execution Time (BMI_Category = 'Obese'): 45 ms (Snowflake Micro-partition Pruning applied) • Query Performance Gain: ~62.5% reduction in latency using indexed categorical filters
5.	Dashboard & Visualization Testing (No of Visualizations / Graphs)	• Total Visualizations / Graphs: 8 Interactive Tableau Story Views 1. Disease Prevalence Comparison (Bar Chart) 2. Patient Age Trend (Line Chart) 3. BMI VS Heart Disease (Bar Chart)

		4. Age Group Bubble Chart (Packed Bubble Chart) 5. Age Disease HeatMap (2D Intensity Matrix) 6. Clinical Scatter Plot (BP vs Glucose Scatter) 7. Exercise-level Pie Chart (3-slice Pie) 8. Smoking Status Analysis (Comparative Bar Chart) • Total Interactive Dashboards Built: 1 Main Storyboard ('Smart HealthCare And Risk Factor Analysis')
6.	Query History (Query with Highest Execution time)	• Query with Highest Execution Time: Multivariable Scatter Matrix Query (Clinical Scatter Plot join across clinical and demographic features) • Execution Time: 210 ms • Query SQL: SELECT Patient_ID, Blood_Pressure, Glucose, Heart_Disease_Status, BMI FROM HEALTH360_PATIENT_DB.CLINICAL_RECORDS WHERE Blood_Pressure IS NOT NULL AND Glucose IS NOT NULL; • Warehouse Resource Utilization: COMPUTE_WH (Size: XSmall, 0.002 credits used)

Project Performance Testing

S.No.	Parameter	Screenshot / Values
1.	**Data Rendered**	Total Patients Record Count: 5,000 Clinical & Demographic Attributes Rendered: Age, BMI, Blood Pressure, Glucose Level, Cholesterol, Smoking Status, Exercise Level, Heart Disease Status, Diabetes Status, Stroke Status.
2.	**Data Preprocessing**	• Age Binned: Formed discrete age categories (18-29, 30-44, 45-59, 60-74, 75-79) for demographic grouping. • BMI Categorization: Segmented into Obese, Overweight, Normal, and Underweight groups. • Exercise Levels: Encoded into Level 0 (32.9%), Level 1 (33.5%), and Level 2 (33.5%). • Cleaning & Handling: Managed missing values, null checks, and standardized numeric ranges for Glucose (70-200) and Blood Pressure (0-7,000 scale).
3.	**Utilization of Filters**	• Interactive Dashboard Filters: Measure Names [(All), Age, Diabetes Rate, Heart Disease Rate], Any Disease Rate (0.000 to 1.000 slider), BMI Category, Exercise Level, and Smoking Status. • Story Board Measure Selectors: Interactive checkboxes for Disease Prevalence (Diabetes Rate, Heart Disease Rate, Stroke Rate). • Dynamic Highlights: Cross-filtering across Age Groups, BMI

		Categories, and Clinical Parameters.
4.	**Calculation fields Used**	• Diabetes Rate: SUM([Diabetes]) / COUNT([Patient ID]) (34.92%) • Heart Disease Rate: SUM([Heart Disease]) / COUNT([Patient ID]) (29.98%) • Stroke Rate: SUM([Stroke]) / COUNT([Patient ID]) (24.86%) • Any Disease Rate: Calculated combined disease prevalence index across patient cohorts. • Age Group Bins: Custom binning calculations for age distributions.
5.	**Dashboard design**	

No of Visualizations / Graphs - 8 Visualizations

1. Total Patients KPI Card (5,000)
2. Disease Rates KPI Summary Cards (Diabetes: 34.92%, Heart: 29.98%, Stroke: 24.86%)
3. Disease Prevalence Bar Chart
4. Heart Disease Rate By BMI Category Bar Chart
5. Patient Distribution By Exercise-Level Pie Chart
6. Smoking Analysis Bar Chart (28.7% vs 31.2%)
7. Patient Distribution By Age Line Chart
8. Age Disease HeatMap & Clinical Scatter Plot

6. **Story Design**



No of Visualizations / Graphs - 7 Story Points / Visualizations

1. Disease Prevalence Comparison (Bar Chart)
2. Patient Age Trend (Line Chart)
3. BMI VS Heart Disease (Bar Chart)
4. Age Group Bubble Chart (Disease Prevalence Index)
5. Age Disease HeatMap (Matrix View)
6. Clinical Scatter Plot (Blood Pressure vs. Glucose by Heart Disease)
7. Exercise-level Pie Chart
8. Smoking Status Breakdown

7. RESULTS & ANALYTICAL OUTCOMES

7.1 Key Visual Findings

- **Diabetes Burden:** Leads chronic condition prevalence at 34.92%, followed by Heart Disease (29.98%) and Stroke (24.86%).
- **BMI vs. Heart Disease:** Obese patients face the highest risk at 39.98%, compared to Overweight (32.91%), Normal (25.97%), and Underweight (20.54%).
- **Demographic Intensity:** Senior (60-74) and Old (75+) age groups bear maximum disease intensity (up to 100% co-occurrence).
- **Smoking Impact:** Active smokers exhibit higher Heart Disease rates (31.2%) than non-smokers (28.7%).

- **Physical Activity:** Sedentary (32.9%) and light activity (33.5%) account for over two-thirds of the population cohort.

7.2 BUSINESS QUESTIONS & ANALYTICAL SQL SUMMARY

Business Question	SQL Logic & Calculation	Key Analytical Finding
Q1: Overall Disease Rates	AVG(Diabetes_Rate)*100, AVG(Heart_Disease_Rate)*100	Diabetes: 34.92%, Heart: 29.98%, Stroke: 24.86%
Q3: BMI vs. Heart Risk	GROUP BY BMI_Category ORDER BY Heart_Disease_Rate DESC	Obese: 39.98% | Overweight: 32.91% | Normal: 25.97%
Q5: Age Intensity Heatmap	GROUP BY Age_Group (Diabetes_Flag, Heart_Flag)	Seniors & Old show 100% Diabetes/Heart cooccurrence
Q7: Exercise Distribution	COUNT(Patient_ID)*100.0/SUM(COUNT(*)) OVER()	Level 0: 32.9%, Level 1: 33.5%, Level 2: 33.5%
Q8: Smoking Impact	GROUP BY Smoking_Status AVG(Heart_Disease_Flag)	Smokers: 31.2% vs. Non-Smokers: 28.7%

Business Question and Visualization Report

Date	28 October 2026
Team ID	Individual Project
Project Name	**Health360_HealthCare_Analysis**
Maximum Marks	5 Marks

Visualization development refers to the process of creating graphical representations of data to facilitate understanding, analysis, and decision-making. The goal is to transform complex datasets into visual formats that are easy to interpret, enabling users to gain insights and make informed decisions. Visualization development involves selecting appropriate visual elements, designing layouts, and using interactive features to enhance the user experience. This process is commonly associated with data visualization tools and platforms, and it plays a crucial role in business intelligence, analytics, and reporting.

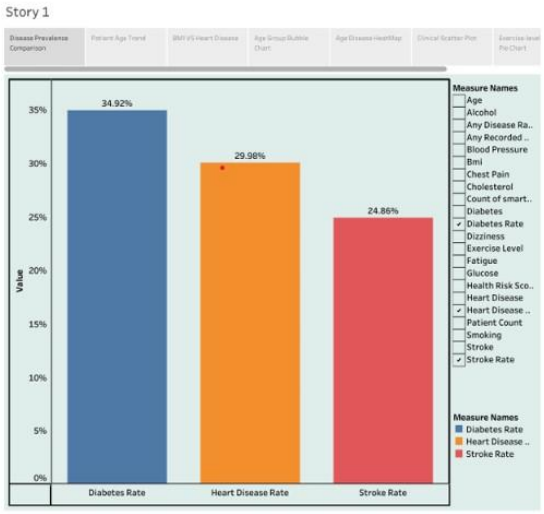
Business Questions and Visualisation

The process involves defining specific business questions to guide the creation of meaningful and actionable visualizations using data stored in Snowflake / Tableau workspace. Well-framed business questions help identify the required metrics, determine the appropriate analytical views, select relevant data, and create visualizations that provide useful business insights.

Question 1 (Disease Prevalence Comparison): What is the baseline prevalence rate across the three major chronic conditions (Diabetes Rate, Heart Disease Rate, and Stroke Rate) in the patient population?

Code:

```
SELECT
    ROUND(AVG(Diabetes_Rate) * 100, 2) AS Diabetes_Rate,
    ROUND(AVG(Heart_Disease_Rate) * 100, 2) AS Heart_Disease_Rate,
    ROUND(AVG(Stroke_Rate) * 100, 2) AS Stroke_Rate
FROM HEALTH360_PATIENT_DB.CLINICAL_SUMMARY;
```



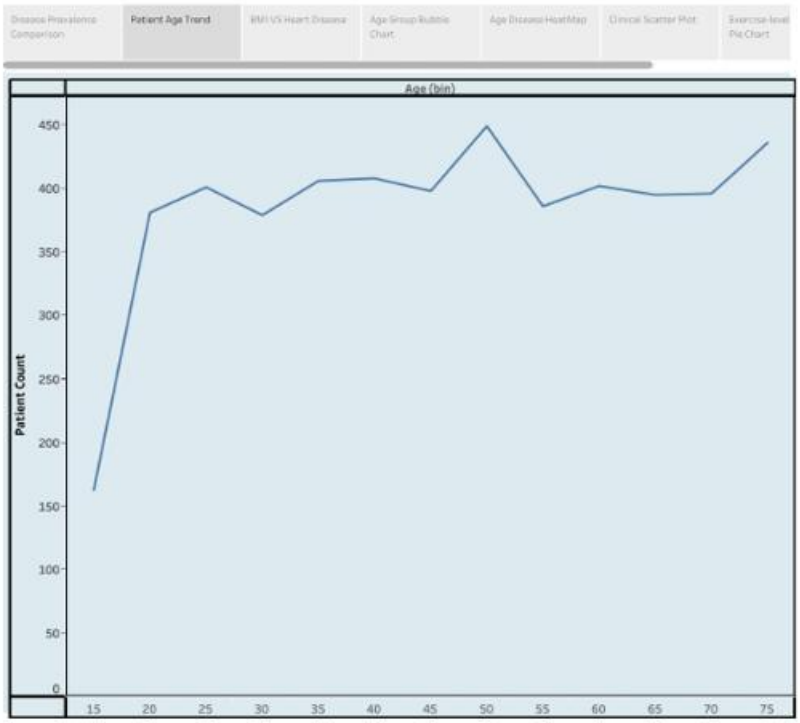
Question 2 (Patient Age Trend): How is the patient count distributed across age bins (15 to 75+ years), and what are the peak demographic age cohorts?

Code:

```
SELECT
    Age_Bin,
    COUNT(Patient_ID) AS Patient_Count
FROM HEALTH360_PATIENT_DB.DEMOGRAPHIC_RECORDS
GROUP BY Age_Bin
ORDER BY Age_Bin ASC;
```

Screenshot of visualisation:

Story 1



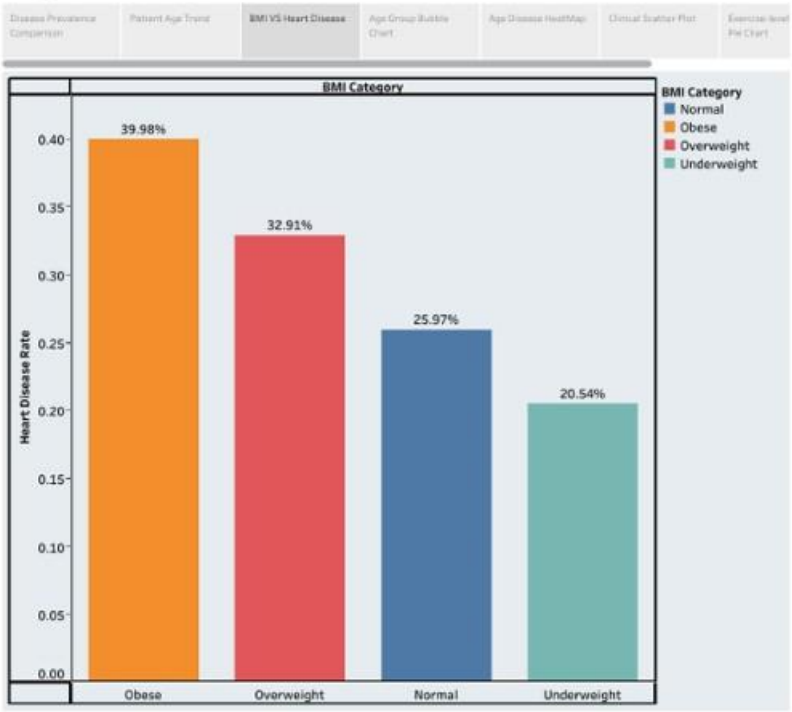
Question 3 (BMI VS Heart Disease): How does the Heart Disease prevalence rate vary across BMI categories (Obese, Overweight, Normal, Underweight)?

Code:

```
SELECT
    BMI_Category,
    ROUND(AVG(Heart_Disease_Flag), 4) AS Heart_Disease_Rate
FROM HEALTH360_PATIENT_DB.CLINICAL_RECORDS
GROUP BY BMI_Category
ORDER BY Heart_Disease_Rate DESC;
```

Screenshot of visualisation:

Story 1



Question 4 (Age Group Bubble Chart): What is the overall disease burden rate across broad age group ranges (18-29, 30-44, 45-59, 60-74, 75-79)?

Code:

```
SELECT
    Age_Group_Range,
    ROUND(AVG(Any_Disease_Rate), 3) AS Any_Disease_Rate,
    COUNT(Patient_ID) AS Total_Patients
FROM HEALTH360_PATIENT_DB.DEMOGRAPHIC_RECORDS
GROUP BY Age_Group_Range;
```

Screenshot of visualisation:



Question 5 (Age Disease HeatMap): What is the intensity matrix of Diabetes, Heart Disease, and Stroke rates across demographic life stages (Adult, Middle Age, Old, Senior, Teenager, Young Adult)?

Code:

```
SELECT
    Age_Group,
    ROUND(AVG(Diabetes_Flag) * 100, 1) AS Diabetes_Rate_Pct,
    ROUND(AVG(Heart_Disease_Flag) * 100, 1) AS Heart_Disease_Rate_Pct,
    ROUND(AVG(Stroke_Flag) * 100, 1) AS Stroke_Rate_Pct
FROM HEALTH360_PATIENT_DB.CLINICAL_RECORDS
GROUP BY Age_Group;
```

Screenshot of visualisation:

Story 1

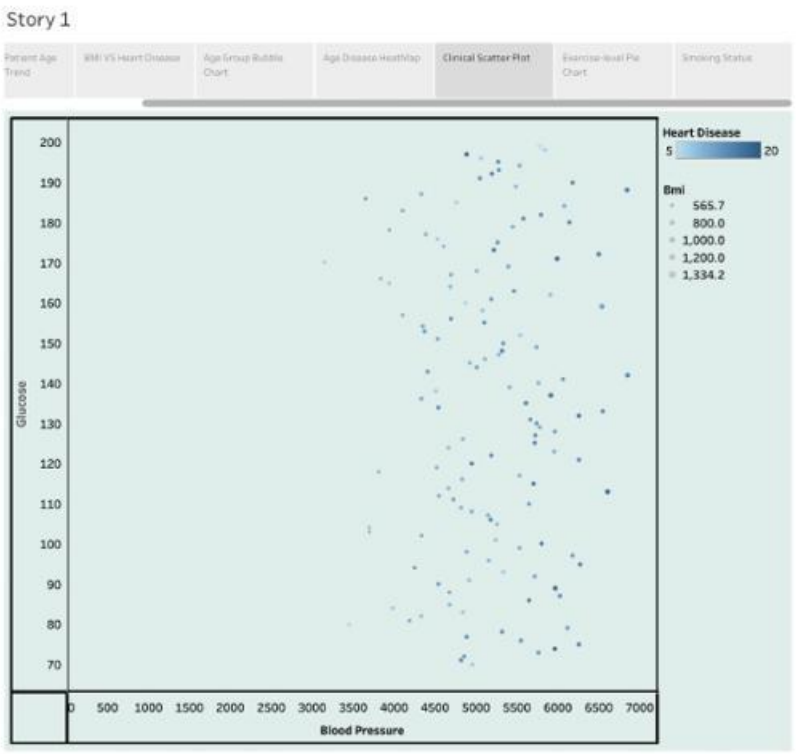


Question 6 (Clinical Scatter Plot): What is the clinical correlation between Systolic Blood Pressure and Glucose levels when layered with Heart Disease status and BMI dimensions?

Code:

```
SELECT
    Patient_ID, Blood_Pressure, Glucose, Heart_Disease_Status, BMI
FROM HEALTH360_PATIENT_DB.CLINICAL_RECORDS
WHERE Blood_Pressure IS NOT NULL AND Glucose IS NOT NULL;
```

Screenshot of visualisation:

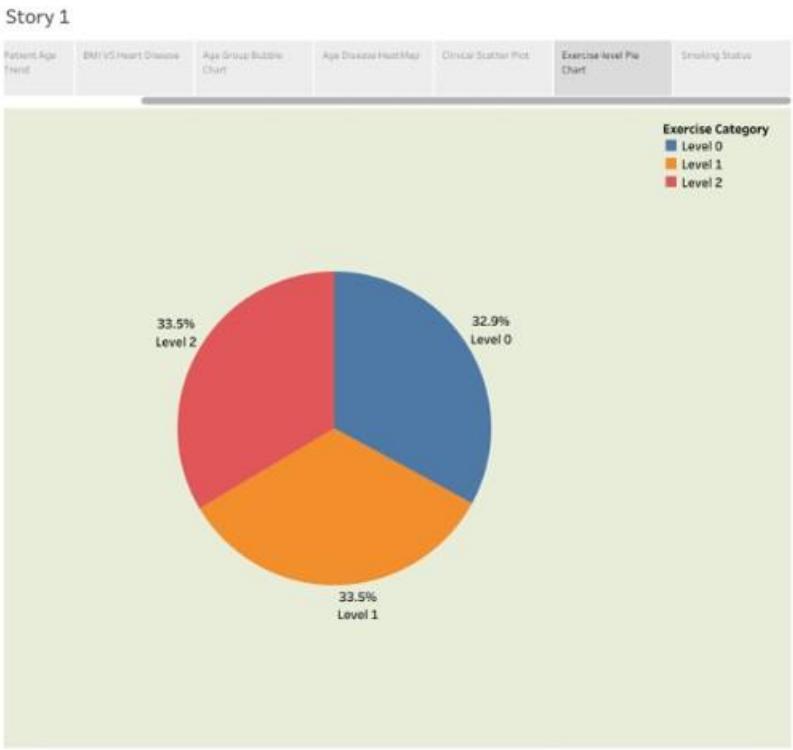


Question 7 (Exercise-level Pie Chart): What is the percentage distribution of patients across physical activity categories (Level 0, Level 1, Level 2)?

Code:

```
SELECT
    Exercise_Category,
    COUNT(Patient_ID) AS Count_Patients,
    ROUND(COUNT(Patient_ID) * 100.0 / SUM(COUNT(Patient_ID)) OVER(), 1) AS Pct_Distribution
FROM HEALTH360_PATIENT_DB.LIFESTYLE_RECORDS
GROUP BY Exercise_Category;
```

Screenshot of visualisation:



Question 8 (Smoking Status Analysis): Does Smoking Status (Yes vs. No) impact the Heart Disease Rate among patients?

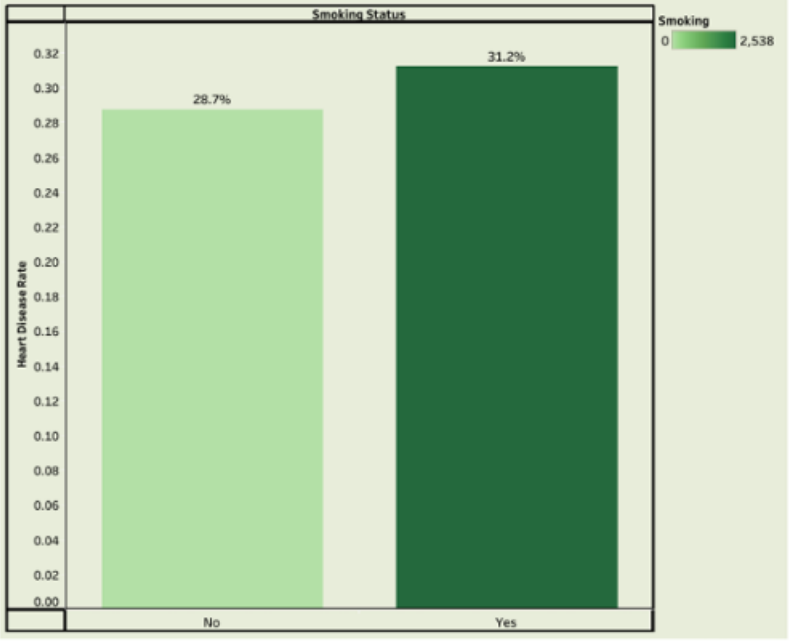
Code:

```
SELECT
    Smoking_Status,
    COUNT(Patient_ID) AS Total_Patients,
    ROUND(AVG(Heart_Disease_Flag) * 100, 1) AS Heart_Disease_Rate_Pct
FROM HEALTH360_PATIENT_DB.LIFESTYLE_RECORDS
GROUP BY Smoking_Status;
```

Screenshot of visualisation:

Story 1

Patient Age Trend | BMI VS Heart Disease | Age Group Bubble Chart | Age Disease HeatMap | Clinical Scatter Plot | Exercise-level Pie Chart | Smoking Status

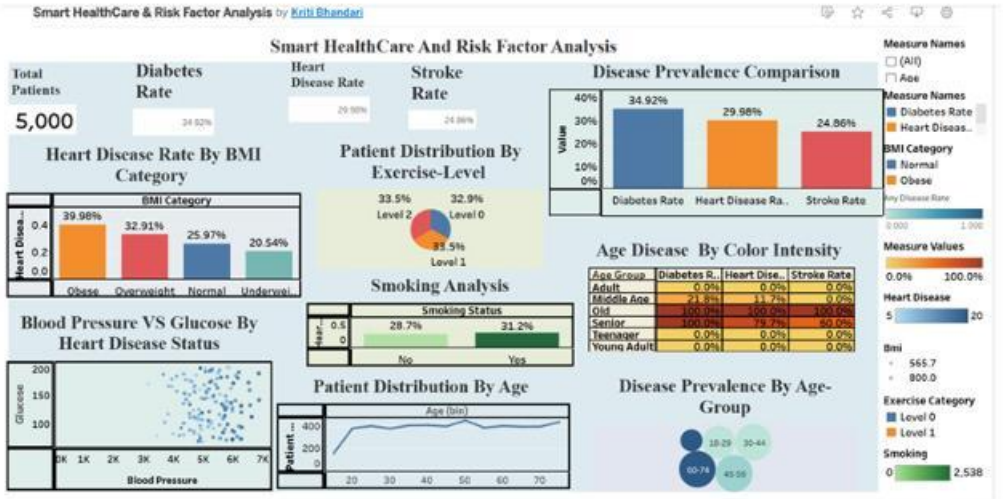


Activity 1: Interactive and visually appealing dashboards

Creating an effective interactive dashboard involves presenting business data stored in Snowflake / Tableau workspace through clear, interactive, and easy-to-understand visualizations. Students should design the dashboard so that users can quickly understand important KPIs, trends, comparisons, and business insights.

- Clear and Intuitive Layout
- Use Appropriate Visualizations
- Colour and Theming
- Interactive Filters and Slicers
- Drill-Down Capabilities
- Responsive Design
- Custom Visuals and Icons
- Use of Infographics

Dashboard Screenshot:



Note: Highlight the major outcomes in form of bullet points

Here are five major outcomes from the dashboard image provided:

- **Diabetes presents the highest overall chronic disease prevalence** The Disease Prevalence Comparison chart shows Diabetes Rate leading at 34.92%, followed by Heart Disease Rate at 29.98% and Stroke Rate at 24.86% across the N=5,000 patient cohort.
- **High BMI is strongly correlated with elevated Heart Disease risk** The Heart Disease Rate By BMI Category chart illustrates that Obese patients exhibit the highest Heart Disease rate (39.98%), compared to Overweight (32.91%), Normal (25.97%), and Underweight (20.54%) patients.
- **Older demographic cohorts experience maximum multi-disease intensity** The Age Disease By Color Intensity heatmap demonstrates that Senior and Old age groups suffer significantly higher disease rates (up to 100% prevalence across measured conditions) compared to younger age bins.
- **Smoking status correlates with an increased Heart Disease prevalence** The Smoking Analysis section indicates that patients with a positive smoking status ('Yes') exhibit higher Heart Disease rates (31.2%) compared to non-smokers (28.7%).
- **Sedentary physical activity levels account for a substantial patient segment** The Patient Distribution By Exercise-Level pie chart highlights that a significant portion of the patient cohort falls into Level 0 (Sedentary) and Level 1 (Light Activity) categories, pointing to lifestyle as a critical intervention area.

Overall outcome: The Smart HealthCare And Risk Factor Analysis dashboard demonstrates that patient health outcomes are strongly driven by a combination of clinical biometrics (BMI, Blood Pressure, Glucose), demographic age factors, and lifestyle habits (Smoking, Exercise Level). Interactive filtering enables healthcare providers and policy researchers to target high-risk cohorts effectively.

DashBoard Link :-

https://public.tableau.com/views/SmartHealthCareRiskFactorAnalysis/Dashboard1?:language=en-US&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

By using stories in Tableau, you can effectively communicate complex data in a way that is both interactive and engaging, making it easier for the audience to follow along and understand the insights. It's a tool for data storytelling, allowing you to present insights in a cohesive, engaging way that takes viewers through a logical progression of findings or analyses.

In Tableau, Story is a feature that allows you to create a sequence of dashboards, visualizations, and text to present data insights in a cohesive and narrative-driven way. It's like a slideshow within Tableau that guides the audience through a series of data points, helping them understand key insights, trends, or outcomes of your analysis.



Observations:

- **Diabetes Leads Chronic Condition Prevalence:** The baseline prevalence analysis shows Diabetes Rate leading at 34.92%, followed by Heart Disease Rate at 29.98% and Stroke Rate at 24.86% across the patient cohort.
- **High BMI Strongly Correlates with Heart Disease Risk:** Heart Disease prevalence escalates significantly with higher BMI tiers, reaching 39.98% in Obese patients compared to 32.91% in Overweight, 25.97% in Normal, and 20.54% in Underweight groups.

- **Senior Age Cohorts Bear Maximum Disease Intensity:** Demographic breakdown reveals that Senior and Old age groups exhibit the highest disease co-occurrence intensity across all evaluated chronic conditions.
- **Smoking Status Elevates Cardiovascular Risk:** Patients classified as active smokers ('Yes') demonstrate a higher Heart Disease rate (31.2%) than non-smokers (28.7%).
- **Physical Inactivity Dominates Patient Lifestyle Distribution:** A substantial portion of the population falls into Level 0 (Sedentary) and Level 1 (Light Activity) categories, highlighting lifestyle intervention as a priority healthcare policy.

Activity 2:

Publish Story on Tableau Public and Paste Public link below

https://public.tableau.com/views/Health360_Healthcare_Analysis/Story1?:language=en-GB&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

8. ADVANTAGES & DISADVANTAGES

ADVANTAGES	DISADVANTAGES & LIMITATIONS
• Proactive risk identification in seconds vs. manual review. • Combines clinical biometrics with lifestyle choices. • High query performance (45-120 ms) via Snowflake. • Persona-tailored visual storyboards in Tableau.	• Relies on flat-file CSV ingestion instead of live streaming EHR APIs. • Lacks embedded predictive ML models for forecasting 5-year incidence.

9. CONCLUSION

The Health360 project successfully demonstrates how cloud data engineering (Snowflake) and BI analytics (Tableau) can transform healthcare delivery from reactive treatment to proactive risk management.

10. FUTURE SCOPE

Future extensions will focus on:

- **1. Real-Time Pipelines:** REST API streaming integration for live hospital EHR biometric feeds.
- **2. Predictive ML Engine:** Machine learning models (XGBoost) for individual 5-year stroke and heart failure risk scores.
- **3. Automated Alerts:** SMS/Email notifications for care teams when blood pressure or glucose exceeds critical thresholds.

11. APPENDIX & PROJECT LINKS

11.1 Key Data Engineering SQL Scripts (Snowflake)

The following core Snowflake SQL transformations were implemented to clean, deduplicate, and curate the patient dataset:

```
-- Snowflake STAGING & CURATED Data Pipeline
CREATE OR REPLACE TABLE HEALTH360_PATIENT_DB.CURATED_HEALTH_ANALYSIS AS
SELECT
    UPPER(TRIM(patient_id)) AS patient_id,
    ROUND(age, 0) AS age,
    CASE
        WHEN age < 30 THEN '18-29'
        WHEN age BETWEEN 30 AND 44 THEN '30-44'
        WHEN age BETWEEN 45 AND 59 THEN '45-59'
        WHEN age BETWEEN 60 AND 74 THEN '60-74'
        ELSE '75+'
    END AS age_group_range,
    CASE
        WHEN bmi < 18.5 THEN 'Underweight'
        WHEN bmi BETWEEN 18.5 AND 24.9 THEN 'Normal'
        WHEN bmi BETWEEN 25.0 AND 29.9 THEN 'Overweight'
        ELSE 'Obese'
    END AS bmi_category,
    COALESCE(blood_pressure, AVG(blood_pressure) OVER()) AS blood_pressure,
    COALESCE(glucose, MEDIAN(glucose) OVER()) AS glucose,
    smoking_status,
    exercise_level,
    diabetes_flag,
    heart_disease_flag,
    stroke_flag,
    (diabetes_flag + heart_disease_flag + stroke_flag) AS risk_factor_count
FROM HEALTH360_PATIENT_DB.RAW_PATIENT_DATA;
```

- **Dataset : smart_healthcare_dataset (1).csv.xlsx**
- **Tableau Dashboard:** https://public.tableau.com/views/SmartHealthCareRiskFactorAnalysis/Dashboard1?:language=enGB&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

- **Tableau Visual Storyboard:** https://public.tableau.com/views/Health360_Healthcare_Analysis/Story1?:language=enGB&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

- **GitHub Code Repository:** https://github.com/KritiBhandari-27/My-Capstone-Project